

Technology Stack

Date	27 June 2026
Team ID	ghousiyaunnissa@gmail.com
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

S.NO	ARCHITECTURE COMPONENT / LAYER	TECHNOLOGY CHOSEN	JUSTIFICATION / PURPOSE
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js / HTML5 / CSS3	Provides a highly responsive, modern user interface with reusable component structures. It allows users to input financial parameters and receive instantaneous prediction results via asynchronous API communication.
2	Backend / Server-Side (e.g., Node.js, Python/ Django, Java)	Python / FastAPI	Chosen for its high performance and seamless integration with Python-based machine learning pipelines. FastAPI provides automatic documentation (Swagger UI) and asynchronous capabilities for fast inference speeds.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	PostgreSQL	A robust, relational database that ensures transactional integrity (ACID compliance) for structured user profile data, logs, and prediction histories. Ideal for complex relational mapping of customer records.

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4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Docker & AWS (EC2 / ECS)	Docker ensures environment consistency across development and production. AWS provides reliable cloud infrastructure, horizontal auto-scaling, and secure hosting for both the frontend and the ML service.
5	Version Control & CI/ CD (e.g., GitHub, GitLab, Jenkins)	GitHub & GitHub Actions	Enables seamless team collaboration, version history tracking, and an automated continuous integration pipeline to test backend models and UI code before automated deployment.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Scikit-learn / Pandas / N umP y	Core data science libraries used to handle feature engineering, data preprocessing, and to serve the trained predictive Machine Learning models for credit scoring assessment.